Lesson 1: Money Sense: Rounding, Estimating and Using Your Calculator

Reading money amounts, rounding them, estimating a bill in your head, and using the TI-30XIIS for exact totals and change.

1. Place Value with Money

Money is written with a **decimal point**. The digits to the left of the point are whole dollars. The digits to the right are parts of a dollar.

- The first place after the point is **tenths**: dimes. \$0.30 is 3 dimes, or 30 cents.
- The second place is **hundredths**: cents. \$0.07 is 7 cents.
- The third place is **thousandths**: tenths of a cent. You can't pay a tenth of a cent, but you see it on gas prices (\$1.479 per litre) and on shelf tags that compare prices.

\$1 = 100 cents 1 dime = \$0.10 1 cent = \$0.01 a tenth of a cent = \$0.001	
Cents to dollars: divide by 100 (move the decimal point 2 places left).	85¢ = \$0.85 149.9¢ = \$1.499

Watch out. A calculator shows \$4.50 as `4.5`. That is four dollars and *fifty* cents, not \$4.05. Always write money with two decimal places: \$4.50.

Example 1

(a) Write each digit of \$23.586 in the place-value chart.

Tens (\$10 bills)	**Ones** (dollars)		**Tenths** (dimes)	**Hundredths** (cents)	**Thousandths** (tenths of a cent)
		.			

(b) What is the value of the 8 in \$23.586? _____

(c) Write each amount in dollars.

(i) six dollars and four cents _____ (ii) 75¢ _____ (iii) 139.9¢ per litre _____

(d) Put these prices in order from least to greatest: \$2.5 \$2.05 \$2.45 \$2.54

2. Rounding Money

A calculator often shows more decimal places than you can pay. Then you **round**: you replace the number with a nearby number that is easier to use.

To round: **Step 1.** Find the place you are rounding to. **Step 2.** Look at the digit just to its *right*.
Step 3. If that digit is **5 or more**, round up (add 1 to your place). If it is **4 or less**, keep your place the same. Drop the digits after it.

To the nearest **dollar**:	look at the tenths (dimes) digit.
To the nearest **ten cents**:	look at the hundredths (cents) digit.
To the nearest **cent**:	look at the thousandths digit.

Watch out. When you round up a 9, it becomes 10 and you carry 1 to the next place: \$6.97 to the nearest ten cents is \$7.00, not "\$6.10". Also, round only at the *end* of a calculation. Rounding in the middle can make your final answer a few cents off.

Example 2

(a) Round each amount.

Amount	Nearest dollar	Nearest ten cents	Nearest cent
\$8.274			
\$15.639			
\$0.958			
\$29.996			

(b) Kim puts 42 L of gas in her car. Gas costs \$1.479 per litre. Her calculator shows `62.118`. How much does she pay?

3. Estimating a Bill

An **estimate** is an answer that is close to the exact answer and that you can find in your head. Estimating tells you if you have enough money, and if a calculator or a receipt answer makes sense.

Rounding estimate: round each price to the nearest dollar, then add.
Front-end estimate: add the dollar parts first (the "front" digits). Then look at the cents and group them into amounts of about \$1. Add those dollars on.

Tip. Need to be *sure* you have enough money? Round every price *up* to the next dollar (\$3.20 becomes \$4). That estimate is a bit too high, so if you have that much, you have enough.

Example 3

Jordan's grocery list is shown.

Item	Price
Milk, 4 L	\$6.49
Loaf of bread	\$3.29
Bananas	\$1.87
Peanut butter	\$5.79
Cereal	\$4.99

(a) Find a front-end estimate of the total.

(b) Find a rounding estimate of the total.

(c) Find the exact total with your calculator.

(d) Jordan has a \$20 bill and a \$5 bill. Is that enough?

Is the answer reasonable? Estimate first, then calculate. If the two answers are far apart, check what you typed. A decimal point in the wrong place makes an answer 10 or 100 times too big or too small.

Example 4

(a) Emma and a friend each order a burger at \$16.95 and a milkshake at \$6.25. The bill says \$56.40. Estimate the bill. Does it make sense?

(b) Tyler buys 5 bags of chips at \$3.89 each. His calculator shows `194.5`. Is this reasonable? What did he probably do wrong?

4. Order of Operations on the Calculator

When a calculation has more than one operation, you must do them in a set order, called the **order of operations**. Remember it with **BEDMAS**.

Brackets first → **E**xponents → **D**ivision and **M**ultiplication (left to right) → **A**ddition and **S**ubtraction (left to right)

The TI-30XIIS follows BEDMAS by itself, so you can type a whole line and it multiplies before it adds. When you need to *add first* and then divide or multiply, put the adding in brackets.

Calculator. Type the whole calculation on one line, then press [ENTER] once at the end. Use [(] and [)] for brackets. The top line of the screen shows what you typed: check it before you press [ENTER]. To start over, press [CLEAR].

Example 5

(a) Work out each by hand. (i) $4 + 6 \times 2$ (ii) $(4 + 6) \times 2$

(b) Nadia buys 3 jars of salsa at \$4.29 each and 2 bags of tortilla chips at \$1.89 each. Write one calculation for the total, find it with your calculator, and check it with an estimate.

Watch out. To share a bill, you need the *total* first. Put the adding in brackets: $(a + b + c) \div 4$. Without brackets, the calculator divides only the last number.

Example 6

Four coworkers share a lunch order: a pizza for \$28.99, wings for \$16.49 and drinks for \$6.52. They split the cost equally.

(a) Write one calculation with brackets and find each person's share.

(b) Sam types 28.99 [+] 16.49 [+] 6.52 [÷] 4 [ENTER] and gets `47.11`. Explain what went wrong.

5. Multiplying, Dividing and Making Change

Multiply to find the cost of many of the same item, or your pay for many hours (wage × hours). **Divide** to share a cost, or to find the price of one item in a pack. Estimate first, use the calculator, then round to the nearest cent at the end.

Example 7

(a) Diego works at a garden centre for $16.25 per hour. Last week he worked 18.5 hours. How much did he earn?

(b) A case of 24 bottles of water costs $5.99. What does one bottle cost?

(c) Three friends share a $25.00 taxi ride. The calculator shows `8.333333333`. How much should each friend pay?

When you pay with cash, the cashier gives you **change**: the amount you paid minus the price. Cashiers often find change by **counting up** from the price.

Counting up: start at the price. Add small coins to reach a friendly amount (one that ends in 00, 25, 50 or 75 cents), then the next dollar, then add bills up to the amount paid. The change is everything you added.
Check: amount paid − price = change.

No pennies. Canada stopped making pennies in 2013. When you pay with *cash*, the total is rounded to the nearest 5 cents. A total ending in 1 or 2 cents rounds down to 0; 3 or 4 rounds up to 5; 6 or 7 rounds down to 5; 8 or 9 rounds up to the next 10. Card and debit payments are *not* rounded.

Example 8

(a) Lunch costs $13.65. You pay with a $20 bill. Count up to find your change.

Start at	Add	Now at
$13.65		

(b) Your total at a convenience store is $8.42.

(i) What do you pay in cash? _____ (ii) What is your cash change from $10? _____

(iii) What do you pay by debit? _____

Lesson 2: Fractions Refresher: Equivalent Fractions and Simplest Form

What a fraction means, equivalent fractions, the greatest common factor and simplest form, comparing fractions, fractions and decimals on the calculator, and a fraction of an amount.

1. What a Fraction Means

A **fraction** names part of something. It has two numbers.

- The **denominator** (bottom number) tells how many *equal* parts make one whole.
- The **numerator** (top number) tells how many of those parts you have.

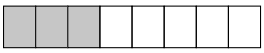
A fraction can be **part of a whole** (3 of the 8 slices of one pizza) or **part of a set** (3 of the 8 people on a work crew).

$$\text{fraction} = \frac{\text{numerator}}{\text{denominator}} = \frac{\text{number of parts you have}}{\text{number of equal parts in one whole (or items in the set)}}$$

Watch out. The parts of a whole must all be the same size. A pizza cut into 4 uneven pieces is not cut into quarters.

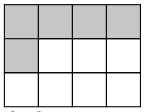
Example 1

Write the fraction of each picture that is shaded, and the fraction that is not shaded.

(a) 

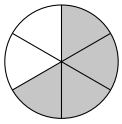
Shaded: _____

Not shaded: _____

(b) 

Shaded: _____

Not shaded: _____

(c) 

Shaded: _____

Not shaded: _____

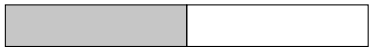
(d) 

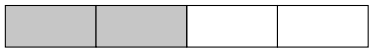
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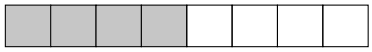
Not filled: _____

2. Equivalent Fractions

Equivalent fractions name the same amount, even though the numbers are different. Each bar below has the same amount shaded, so $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$.

$\frac{1}{2}$ 

$\frac{2}{4}$ 

$\frac{4}{8}$ 

To make an equivalent fraction, multiply *or* divide the numerator and the denominator by the **same** number.

$$\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12} \qquad \frac{15}{20} = \frac{15 \div 5}{20 \div 5} = \frac{3}{4}$$

Watch out. Adding the same number to the top and bottom does *not* work: $\frac{1}{2}$ is not equal to $\frac{1+1}{2+1} = \frac{2}{3}$. Only multiply or divide.

To find a **missing number**, ask: "What was the denominator multiplied (or divided) by?" Then do the same thing to the numerator. Sometimes the jump is not a whole number. Then simplify first (Section 3), and scale from there.

Example 2

(a) Write three fractions equivalent to $\frac{3}{5}$.

(b) Find the missing number: $\frac{2}{3} = \frac{\square}{18}$

(c) Find the missing number: $\frac{20}{32} = \frac{5}{\square}$

(d) Find the missing number: $\frac{4}{7} = \frac{36}{\square}$

Example 3

On a 20-question practice test for his learner's licence, Kenji got 15 questions right. If he does equally well on a 28-question practice test, how many will he get right?

3. Factors, the GCF and Simplest Form

A **factor** of a number divides into it evenly, with no remainder. Find factors in pairs that multiply to the number. For 12: 1×12, 2×6, 3×4, so the factors of 12 are 1, 2, 3, 4, 6, 12.

A **common factor** of two numbers is a factor of both. The **greatest common factor (GCF)** is the largest one. A fraction is in **simplest form** (also called **lowest terms**) when the only common factor of its numerator and denominator is 1.

To write a fraction in simplest form

Step 1: Find the GCF of the numerator and the denominator.
Step 2: Divide the numerator and the denominator by the GCF.

Can't find the GCF? Divide by any common factor you see (2, 3, 5, 10, ...), then look again. Keep going until 1 is the only common factor. You get the same answer; it just takes more steps.

Calculator. The TI-30XIIS simplifies fractions for you. Enter $\frac{18}{24}$ as 18 [A b/c] 24 [ENTER]. The screen shows 3⌟4, which means $\frac{3}{4}$. Use this to check your work, but learn to do it by hand too.

Example 4

(a) List the factors of 24 and 40. Find the GCF, then write $\frac{24}{40}$ in simplest form.

Write each fraction in simplest form.

(b) $\frac{18}{45}$ (c) $\frac{35}{56}$ (d) $\frac{40}{100}$

Example 5

Last week, Rosa, an apprentice electrician, worked 36 hours. She spent 27 of those hours on a job site and the rest in class.

(a) What fraction of her hours was on the job site? Write it in simplest form.

(b) What fraction of her hours was in class? Check it on your calculator.

4. Fractions, Decimals and Comparing Fractions

The fraction bar means **divide**: $\frac{3}{8} = 3 \div 8 = 0.375$. Some fractions make a decimal that **repeats** forever, like $\frac{2}{3} = 0.666\ldots$; round these (to the nearest thousandth, 0.667, is usually enough).

Calculator. Fraction to decimal: enter the fraction, e.g. 3 [A b/c] 8 [ENTER], then press [2nd] [F<>D] to see 0.375. Press [2nd] [F<>D] again to go back. (Or just key 3 [÷] 8 [ENTER].)
Decimal to fraction: key 0.35 [ENTER], then [2nd] [F<>D] shows 7⌟20, which is $\frac{7}{20}$.

Watch out. Divide the top by the bottom. $\frac{3}{8}$ is $3 \div 8 = 0.375$, not $8 \div 3$. A fraction less than one whole always gives a decimal less than 1.

To **compare** two fractions, you need to see which is bigger. Use one of two methods.

Method 1: common denominator. Write both fractions with the same denominator. Then the one with the bigger numerator is bigger.

Method 2: decimals. Change both fractions to decimals and compare the decimals.

Use $<$ ("is less than") and $>$ ("is greater than"). The point of the sign faces the smaller number: $\frac{1}{2} < \frac{3}{4}$.

Watch out. A bigger denominator means *smaller* pieces. $\frac{1}{8}$ of a pizza is less than $\frac{1}{5}$ of the same pizza.

Example 6

Complete the table. Round repeating decimals to the nearest thousandth.

Fraction	$\frac{3}{8}$	$\frac{7}{20}$	$\frac{5}{6}$			
Decimal				0.6	0.125	0.45

Example 7

(a) Use a common denominator to compare $\frac{3}{4}$ and $\frac{5}{6}$. Write $<$ or $>$.

(b) Use decimals to compare $\frac{5}{8}$ and $\frac{7}{11}$. Write $<$ or $>$.

(c) A mechanic's wrenches are marked in inches: $\frac{5}{8}$, $\frac{1}{2}$, $\frac{9}{16}$, $\frac{3}{4}$. Put the sizes in order from smallest to largest.

Example 8

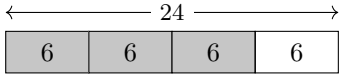
In basketball practice, Ava made 13 of her 20 free throws. Noor made 16 of her 25 free throws.

(a) Who has the better shooting record? Show how you know.

(b) Noor says she is better because she made more shots. Is she right? Explain.

5. A Fraction of an Amount

In math, "of" usually means **multiply**. To find $\frac{3}{4}$ of 24, split 24 into 4 equal groups, then take 3 of them.



$24 \div 4 = 6$ in each part. 3 parts: $3 \times 6 = 18$. So $\frac{3}{4}$ of 24 is 18.

Fraction of an amount Step 1: divide the amount by the denominator. Step 2: multiply by the numerator.

$\frac{3}{4}$ of $24 = 24 \div 4 \times 3 = 18$ Calculator: 24 [×] 3 [÷] 4 [ENTER] or 3 [A b/c] 4 [×] 24 [ENTER]

Is it reasonable? A fraction less than 1 of an amount is always *less* than the amount. $\frac{3}{4}$ is more than half, so $\frac{3}{4}$ of 24 should be more than 12 and less than 24. For money, round to the nearest cent at the end.

Example 9

Find each amount.

(a) $\frac{2}{3}$ of 45 (b) $\frac{5}{8}$ of 64 (c) $\frac{3}{10}$ of \$85 (d) $\frac{2}{7}$ of \$50

Example 10

Dana's car has a 52 L gas tank. The gas gauge shows that the tank is $\frac{1}{4}$ full.

(a) How many litres of gas are in the tank?

(b) How many litres will it take to fill the tank?

(c) Gas costs \$1.45/L. How much will it cost to fill the tank?

Lesson 3: Ratios

Comparing two amounts in the same units: writing, simplifying, scaling up or down, and sharing in a ratio.

1. What Is a Ratio?

A **ratio** compares two amounts that are measured in the **same units**. The numbers in a ratio are called its **terms**. A ratio can be written three ways.

Three ways to write a ratio with a colon $3:2$ in words 3 to 2 as a fraction $\frac{3}{2}$
All three are read "3 to 2". **Order matters:** the first amount named goes first (or on top).

A **part-to-part** ratio compares one part of a group with another part. A **part-to-whole** ratio compares one part with the whole group. To find the whole, add all the parts.

Watch out. $3:2$ and $2:3$ are different ratios. Read the question: "guppies to goldfish" means guppies first. Also, never change a ratio like $\frac{5}{3}$ into a mixed number ($1\frac{2}{3}$). A ratio compares *two* numbers, so keep both.

Example 1

A fish tank at a pet store has 5 goldfish, 3 guppies and 4 tetras. Complete the table.

Comparison	**Colon**	**Words**	**Fraction**	**Part-to-part or part-to-whole?**
goldfish to guppies				
guppies to goldfish				
tetras to all fish				
fish that are not tetras to tetras				

2. Simplifying Ratios

Equivalent ratios are ratios that compare the same way, like $2 : 4$ and $1 : 2$. A ratio is in **simplest form** when its terms are whole numbers with no common factor except 1.

To simplify a ratio: divide both terms by the **greatest common factor** (GCF), the biggest number that divides into both.
$12 : 18 \;\rightarrow\; 12 \div 6 : 18 \div 6 \;\rightarrow\; 2 : 3$ If you can't spot the GCF, divide by any common factor and repeat.

Calculator. The TI-30XIIS simplifies fractions for you. For $12 : 18$, type 12 [A b/c] 18 [ENTER]. The screen shows $2\lrcorner 3$, so the ratio is $2 : 3$. If it shows a mixed number such as $2\mathtt{u}2\lrcorner 3$, press [2nd] [A b/c] (the d/e key) to see $8\lrcorner 3$, which means $8 : 3$.

Example 2

Write each ratio in simplest form.

(a) $12 : 18$

(b) $40 : 15$

(c) $4 : 12$ (the tetras from Example 1)

(d) $24 : 24$

Decimals and different units. The terms of a simplified ratio must be whole numbers, and both amounts must be in the same unit.

Decimals: multiply both terms by 10 or 100 to make whole numbers, then simplify.
Different units: change both amounts to the *same* unit (usually the smaller one), drop the units, then simplify.
1 h = 60 min 1 min = 60 s 1 kg = 1000 g 1 L = 1000 mL 1 m = 100 cm

Example 3

Write each ratio in simplest form.

(a) 2.5 : 4 (b) 0.75 : 1.5 (c) 45 min : 2 h (d) 250 g : 1.5 kg

Example 4

A 2 L jug of iced tea is made with 400 mL of iced tea concentrate. The rest is water.

(a) Write the ratio of concentrate to the whole drink in simplest form.

(b) How much water is in the jug?

(c) Write the ratio of concentrate to water in simplest form.

(d) Which ratio, (a) or (c), is part-to-whole?

3. Equivalent Ratios and Mixing Recipes

Multiply (or divide) **both** terms of a ratio by the **same** number to get an equivalent ratio. A **ratio table** lists equivalent ratios side by side.

Mixing instructions often say "1 part to 4 parts". A **part** can be any size (a cup, a can, a bucket) as long as every part in the mix uses the same size. A ratio can also compare three amounts, like 1 : 2 : 3; multiply or divide all three terms by the same number.

Equivalent ratios: $a : b = (a \times n) : (b \times n)$ e.g. $1 : 4 = 2 : 8 = 3 : 12$
Two ratios are equivalent if they simplify to the same simplest form.

When two ratios are equal, you can write them as an equation called a **proportion**; you will solve these in Lesson 1.4.

Example 5

A can of frozen lemonade concentrate says: mix 1 can of concentrate with 4 cans of water.

(a) Complete the ratio table.

Cans of concentrate	1	2	3		8
Cans of water	4			20	

(b) For a party, Lucas mixes 6 cans of concentrate with 20 cans of water. Is this the same as the instructions? Will his lemonade taste stronger or weaker?

Example 6

A concrete mix for a backyard patio uses cement, sand and gravel in the ratio $1 : 2 : 3$ (by buckets).

(a) Marco uses 4 buckets of cement. How much sand and gravel does he need?

(b) Marco has only 15 buckets of gravel. How much cement and sand go with it?

(c) What is the ratio of cement to the whole mix?

(d) How many buckets of mix does (b) make?

4. Sharing in a Ratio

Sometimes a total (money, a mixture) must be split in a given ratio. Split it into equal **parts** first.

Sharing an amount in the ratio $a : b$
Step 1. Add the terms to get the **total number of parts**: $a + b$.
Step 2. Find **one part**: total amount $\div$ total number of parts.
Step 3. Multiply one part by each term: $a\times$ (one part) and $b\times$ (one part).
Step 4. Check: the shares add up to the total.

Watch out. For $3 : 5$, do not divide by 3 or by 5. Divide by $3 + 5 = 8$, the total number of parts.

Example 7

(a) Jaya and Owen earn $120 shovelling driveways. Owen shovelled more, so they share the money in the ratio 3 : 5 (Jaya : Owen). How much does each get?

(b) Ellie worked 6 h and Ravi worked 9 h at a car wash. They share $187.50 in tips in the ratio of the hours they worked. How much does each get?

Example 8

Windshield washer fluid is mixed 1 part concentrate to 5 parts water. Nadia wants to fill a 4.5 L jug. How much concentrate and how much water does she need?

Lesson 4: Proportions: Finding the Missing Value

Two equal ratios make a proportion: scale up or down, or cross-multiply, to find the missing number in recipes, maps, fuel mixes, doses and photos.

1. What Is a Proportion?

A **proportion** is a statement that two ratios are equal. For example, $2 : 3 = 4 : 6$ is a proportion. Written as fractions, it is $\frac{2}{3} = \frac{4}{6}$.

To decide if two ratios form a proportion, use one of these:

- **Simplify both ratios.** If they simplify to the same ratio, they form a proportion.
- **Divide.** Turn each ratio into a decimal (top ÷ bottom). If the decimals are the same, they form a proportion.

Example 1

Do the two ratios form a proportion? Show how you know.

(a) $4 : 6$ and $10 : 15$ (b) $\frac{3}{8}$ and $\frac{9}{16}$ (c) $12 : 20$ and $9 : 15$

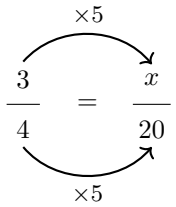
Example 2

Three friends make fruit punch from drink crystals and water. Jordan mixes 2 scoops with 3 L of water. Priya mixes 5 scoops with 7.5 L of water. Marco mixes 3 scoops with 5 L of water.

(a) Do Jordan's and Priya's punches taste the same strength? (b) Is Marco's punch stronger or weaker than Jordan's?

2. Solving by Scaling Up or Down

Often one number in a proportion is missing. We call it x. If a top or bottom number is an easy multiple of its partner, **scale**: find what the known pair was multiplied (or divided) by, then do the same to the other number.



Step 1. Look at the pair with both numbers known (here 4 and 20). What do you multiply or divide by to get from one to the other? ($4 \times 5 = 20$)

Step 2. Do the same thing to the other pair: $x = 3 \times 5 = 15$.

Scaling rule. Multiply or divide the top *and* the bottom of a ratio by the **same** number and you get an equal ratio.

Watch out. Only multiply or divide. Adding the same number to the top and bottom does *not* make an equal ratio: $\frac{3}{4}$ and $\frac{3+16}{4+16} = \frac{19}{20}$ are not equal.

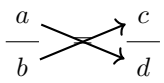
Example 3

Solve each proportion by scaling. Write the $\times$ or $\div$ you used.

(a) $\frac{3}{4} = \frac{x}{20}$ (b) $\frac{5}{8} = \frac{35}{x}$ (c) $\frac{18}{24} = \frac{x}{4}$ (d) $6 : x = 30 : 45$

3. Solving by Cross-Multiplying

When the numbers do not scale nicely (like $\frac{6}{11} = \frac{x}{25}$), **cross-multiply**. In a proportion, the two numbers that sit diagonally across from each other multiply to the same answer. These answers are called the **cross products**.



Cross-multiplying. If $\frac{a}{b} = \frac{c}{d}$, then $a \times d = b \times c$.

Step 1. Multiply across both diagonals and write an equation.
Step 2. Divide both sides by the number that is with x.
Step 3. Check: put your answer in and find both cross products. They should be equal (or very close if you rounded).

For example, $\frac{x}{12} = \frac{7}{8}$ gives $8 \times x = 12 \times 7$, so $8x = 84$ and $x = 84 \div 8 = 10.5$.
Check: $10.5 \times 8 = 84$ and $12 \times 7 = 84$. ✓

Watch out. Multiply along the *diagonals* (top-left with bottom-right, bottom-left with top-right). Do not multiply the two tops together. It works the same way when x is on the bottom.

Calculator. Once you have, say, $11x = 6 \times 25$, you can find x in one line: 6 × 25 ÷ 11 ENTER gives 13.636 36... Round only at the end.

Example 4

Solve by cross-multiplying. Round to the nearest tenth where needed. Check part (b).

(a) $\frac{9}{x} = \frac{15}{40}$ (b) $\frac{6}{11} = \frac{x}{25}$ (c) $\frac{2.5}{4} = \frac{7}{x}$

4. Setting Up a Proportion from a Word Problem

The hardest part of a word problem is writing the proportion. Write it in *words* first, then put in the numbers.

Step 1. Decide which two things are being compared (for example, cups of flour and muffins).
Step 2. Write a word proportion with the **same thing in the same position** on both sides: $\frac{\text{flour}}{\text{muffins}} = \frac{\text{flour}}{\text{muffins}}$.
Step 3. Put in the numbers and use x for the missing one. Units in the same position must match (both mL, or both L). Convert first if they don't.
Step 4. Solve by scaling or cross-multiplying.
Step 5. Ask, "Does my answer make sense?" Then write a sentence answer with units.

Example 5

A muffin recipe uses 2.5 cups of flour to make 12 muffins.

(a) How much flour is needed for 30 muffins? Estimate first.

(b) Lena has only 4 cups of flour. How many whole muffins can she make?

Example 6

On a hiking map, 2 cm represents 5 km of real distance.

(a) Two campsites are 7.4 cm apart on the map. How far apart are they really?

(b) A trail is 12 km long. How long is it on the map?

Example 7

A leaf blower has a 2-stroke engine. Its fuel must be mixed with gas and oil in the ratio 50 : 1 (50 parts gas to 1 part oil).

(a) How many millilitres of oil should be mixed with 4 L of gas?

(b) How much gas, in litres, goes with a full 250 mL bottle of oil?

Watch out. In part (a), writing $\frac{50}{1} = \frac{4}{x}$ (litres on one side, millilitres on the other) gives $x = 0.08$. But 0.08 mL of oil in 4 L of gas makes no sense: it is far too little. Convert to the same units first.

5. Proportions at Work

Nurses, photo technicians and landscapers all use proportions. Set them up the same way. Then think about the answer: sometimes you must **round up** (you can't buy part of a bag) and sometimes **round down** (you can't make part of a muffin).

Example 8

A nurse gives a child a medicine. The order says 1.5 mL for every 10 kg of body mass. The child has a mass of 24 kg. How many millilitres should the child get? *(These numbers are made up for practice. Real doses always come from a doctor or pharmacist.)*

Example 9

A photo is 10 cm wide and 15 cm tall. To enlarge a photo without stretching it, the width and height must stay in the same ratio.

(a) It is enlarged to 24 cm wide. How tall is it?

(b) Can the photo be enlarged to fill a 20 cm by 25 cm frame exactly, without cutting any off? Explain.

Example 10

One bag of grass seed covers 45 m^2 of lawn. Mateo's new lawn is 160 m^2. How many bags must he buy?

Lesson 5: Rates and Unit Rates

Comparing quantities with different units, finding the rate for one unit, and using rates to predict, compare and convert.

1. What Is a Rate?

A **rate** compares two quantities that have **different units**. "90 km in 1 h", "$45 for 3 h of work" and "120 heartbeats in 2 min" are all rates.

A **ratio** (Lesson 3) compares quantities with the **same** unit, like 2 cups of flour to 3 cups of milk. In a rate the units are different, so you must always write them.

You can write the rate "$45 for 3 h" in three ways:

$$\$45 \text{ for } 3 \text{ h} \qquad \frac{\$45}{3 \text{ h}} \qquad \$45/3 \text{ h}$$

The slash "/" is read as **per**. "Per" means "for each" or "for every".

Rate: a comparison of two quantities with **different units**. **Ratio:** a comparison of quantities with the **same unit**.
A rate always keeps its units: 72 pages/3 min, \$6.60/2 kg, 15 L/2 min.

Example 1

Decide whether each statement is a rate or a ratio. Give a reason.

Statement	Rate or ratio?	Reason
(a) A printer prints 72 pages in 3 minutes.		
(b) A recipe uses 2 cups of rice for every 3 cups of water.		
(c) Pears cost \$6.60 for 2 kg.		
(d) A hockey team won 9 games and lost 6 games.		

(e) Write the rate in (a) as a fraction and with a slash. _____ _____

2. Unit Rates

A **unit rate** is a rate for **one** unit of the second quantity. "75 km/h" is a unit rate: it means 75 km in 1 hour. To find a unit rate, divide the first quantity by the second quantity. This is the same as dividing the top and bottom of the rate by the bottom number:

$$\frac{\$45}{3\text{ h}} = \frac{\$45 \div 3}{3\text{ h} \div 3} = \frac{\$15}{1\text{ h}} = \$15/\text{h}$$

Unit rate = first quantity ÷ second quantity		(the answer is the amount for 1 unit)
Speed	km/h	kilometres travelled in 1 hour
Wage (pay)	\$/h	dollars earned in 1 hour
Typing speed	words/min	words typed in 1 minute
Heart rate	beats/min	heartbeats in 1 minute

Watch out. Divide in the right order. The unit after "per" goes on the bottom: dollars *per hour* means dollars ÷ hours. If Jordan earns \$133.20 in 8 h, then $8 \div 133.20 = 0.06$ makes no sense, but $133.20 \div 8 = 16.65$ gives a sensible wage of \$16.65/h.

Example 2

Find each unit rate. Include the units.

(a) A cyclist rides 54 km in 3 h.

(b) Jordan earns \$133.20 for an 8-hour shift at a hardware store.

(c) Mei types 390 words in 6 min.

(d) A tap fills a 24 L bucket in 1.5 min.

Example 3

To find your heart rate, press two fingers on the side of your neck and count the beats. Counting for a whole minute is slow, so people often count for 15 s and multiply by 4 (because $15\text{ s} \times 4 = 60\text{ s}$), or count for 20 s and multiply by 3.

(a) At rest, Aiden counts 18 beats in 15 s. What is his heart rate in beats per minute?

(b) After running up the stairs, he counts 37 beats in 20 s. What is his heart rate now?

3. Using a Unit Rate to Predict

Once you know the amount for 1 unit, you can find the amount for any number of units. You can also work backwards to find how many units (how many hours, minutes or cans) you need.

Amount = unit rate × number of units	**Number of units** = amount ÷ unit rate
You can also set up a **proportion** (Lesson 4). Put the same units in the same places, for example dollars on top and hours on the bottom of both fractions.	

Example 4

Kiran works at a garden centre. He earned \$247.50 for 15 h of work. At this rate, how much will he earn for 22 h? Estimate first, then solve two ways.

Estimate: _____

Method 1: unit rate

Method 2: proportion

Example 5

(a) Talia types 48 words/min. How long will it take her to type a 1200-word essay?

(b) At the deli, smoked turkey costs \$2.38 per 100 g. How much will 450 g cost?

Coming up. A rate that gives a price for an amount, like \$2.38 per 100 g, is called a **unit price**. In Lesson 6 you will use unit prices to find the best buy.

Watch out. The units must match before you multiply or divide. If a rate is per *minute* and the time is in *hours*, change the hours to minutes first (1 h = 60 min). And when you must buy whole items (cans, bags, boxes), **round up**: 4.2 cans means you need 5 cans, because 4 cans are not enough.

Example 6

(a) A machine at a juice plant fills 270 bottles in 45 min. How many bottles can it fill in a 7.5-hour shift?

(b) One can of paint covers 35 m^2. Ty needs to paint 160 m^2 of walls. How many cans should he buy?

4. Fuel Economy: Litres per 100 km

In Canada, the fuel a vehicle uses is given as a rate in **litres per 100 km** (L/100 km). It tells you how many litres of gas the vehicle burns to drive 100 km.

Why per 100 km and not per 1 km? A car burns only a tiny amount of gas in 1 km, such as 0.078 L. Numbers like that are hard to read and compare. For 100 km the same car uses 7.8 L, a friendlier number. A **lower** number means the vehicle uses **less** fuel, which is better.

Fuel consumption (L/100 km) = litres used ÷ km driven ×100
Fuel needed (L) = fuel consumption × km to drive ÷100

Calculator. For 55.2 L used in 480 km, press 55.2 [÷] 480 [×] 100 [ENTER] to get 11.5.

Example 7

On a road trip, Dana's pickup truck used 55.2 L of gas to travel 480 km. What is its fuel consumption in L/100 km?

Example 8

Calgary to Banff is about 130 km each way. Sam's car uses 7.4 L/100 km, and gas costs $1.42/L.

(a) How many litres of gas will Sam need for the round trip (there and back)?

(b) How much will the gas cost?

5. Comparing and Converting Rates

To **compare** two rates, write both as unit rates with the same units. Then compare the numbers. Think about what "better" means: for pay or typing speed, higher is better; for fuel consumption, lower is better.

Example 9

(a) Noah types 504 words in 7 min. Priya types 620 words in 8 min. Who types faster?

(b) An SUV used 46.8 L of gas for 520 km. A hatchback used 31.5 L for 450 km. Which uses less fuel?

To **convert** a rate to new units, change one unit at a time: first the top unit, then the bottom unit.

1 km = 1000 m	1 h = 60 min 1 min = 60 s $1 = 100 cents
km/h to m/min:	×1000 (km to m), then ÷60 (per hour to per minute)
m/min to km/h:	×60 (per minute to per hour), then ÷1000 (m to km)

Example 10

(a) Lena walks at 4.8 km/h. Write her speed in m/min.

(b) An escalator moves at 30 m/min. Write its speed in km/h.

(c) Ethan earns $18.60/h. How much is that per minute?

Lesson 6: Unit Price: Finding the Best Buy

Finding the price of one unit, comparing sizes and deals, and choosing the best buy for cost and for other reasons.

1. What Is a Unit Price?

Stores sell many things in packs, bags and bottles. The **unit price** is the price of **one unit**: one item, one kilogram, one litre, and so on. A unit price is a unit rate with money on top.

Unit price = price ÷ quantity	e.g. 8 batteries for $11.92: $11.92 ÷ 8 = $1.49 per battery

Watch out. Always divide the **price** by the **quantity** (money ÷ amount). If you divide the other way, you get a number that does not make sense. Estimate first: $11.92 for 8 is about $12 for 8, so a bit more than $1 each.

Calculator. 11.92 [÷] 8 [ENTER] shows 1.49. If the screen shows more decimals, round to the nearest cent: 0.74833… is about $0.75.

Example 1

Find each unit price. Round to the nearest cent.

Item	Price	Quantity	Work (price ÷ quantity)	Unit price
(a) Pack of AA batteries	\$11.92	8 batteries		
(b) Bag of potatoes	\$5.97	3 kg		
(c) Bag of bagels	\$4.49	6 bagels		
(d) Jug of milk	\$6.29	4 L		

Example 2

Priya owns a hair salon. She buys a case of 36 bottles of shampoo for \$89.64. She sells each bottle to customers for \$7.50.

(a) How much does one bottle cost Priya?

(b) How much **profit** (money she keeps) does she make on one bottle?

(c) What is her profit if she sells the whole case?

2. Comparing Sizes: Which Is the Best Buy?

To compare two or more sizes, find the unit price of each using the **same unit**. The size with the **lowest unit price** is the **best buy** (when cost is all that matters).

A price per gram or per millilitre is a tiny number, so stores often show the price **per 100 g** or **per 100 mL**. Find it by dividing the price by the number of hundreds: 525 g is 5.25 hundreds.

Price per 100 g = price ÷ grams × 100 (the same for mL: price ÷ mL × 100)

Keep an extra decimal. When you compare unit prices, keep three decimals (to a tenth of a cent), for example \$0.950 and \$0.905. Two prices that round to the same cent can still be different.

Calculator. 4.99 [÷] 525 [×] 100 [ENTER] shows 0.950476…, so about \$0.950 per 100 g.

Example 3

A 525 g box of cereal costs \$4.99. A 750 g box of the same cereal costs \$6.79. Which is the better buy?

(a) Estimate the price per 100 g of each box.

(b) Calculate the price per 100 g of each box, and answer the question.

Watch out. Bigger is **not always** cheaper per unit. Always check.

Example 4

Peanut butter comes in three sizes.

Size	Price	Work	Per 100 g
375 g	\$3.29		
1 kg	\$6.99		
2 kg	\$14.49		

(a) Complete the table. (Remember: 1 kg = 1000 g.)

(b) Which size is the best buy?

_____

(c) You want 2 kg of peanut butter. How much do you save by buying two 1 kg jars instead of one 2 kg jar?

When the units do not match. Sometimes one size is in millilitres and another in litres, or one is in grams and another in kilograms. **Change to the same unit first**, then compare.

1 L = 1000 mL	1 kg = 1000 g	(e.g. 1.5 L = 1500 mL, 650 g = 0.65 kg)

Example 5

(a) A 740 mL bottle of dish soap costs \$3.97. A 1.5 L bottle of the same soap costs \$7.49. Which is the better buy?

(b) At the meat counter, lean ground beef is \$11.00 per kg. A 650 g package in the cooler is \$7.49. Which is the better buy?

3. Multi-Buy Deals and Extra Costs

A **multi-buy deal** gives a price for a group, like "4 for \$5.00". To find the price of one, divide: \$5.00 ÷ 4 = \$1.25 each. Read the sign: some stores give the deal price even if you buy just one, but some say "must buy 4".

If you must pay extra to get the items (**shipping**, delivery), add it to the price **before** you divide.

Unit price with extra costs = (price + shipping or delivery) ÷ quantity

Example 6

At a grocery store, a single cup of yogurt costs \$1.49. The sign also says "4 for \$5.00". A 12-pack of the same cups costs \$13.99.

(a) Find the price of one cup each way. Which is the best buy?

(b) How much do you save by buying the 12-pack instead of 12 single cups?

Example 7

Coach Okafor is ordering water bottles for her soccer club online.

- Store A: 24 bottles for $62.40, plus $9.99 shipping.
- Store B: 30 bottles for $84.00, with free shipping.

(a) Find the price per bottle at Store A *without* shipping.

(b) Find the real price per bottle at each store, with shipping. Which store is the better buy?

4. The Best Buy Is Not Always the Cheapest

The lowest unit price only tells you about **cost**. The best buy for *you* also depends on other things:

- **Will it spoil?** Food you throw out is money wasted.
- **Do you need that much?** A big pack is not a deal if you only use a little.
- **Do you have room to store it?**
- **Quality:** is it the same product, brand and quality?
- **Money today:** can you afford the higher total price right now?

Example 8

Jasmine lives on her own. She eats about 100 g of spinach a week. Fresh spinach stays good for about 5 days. Her store sells a 142 g box for $3.49 and a 1 kg tub for $10.99.

(a) Find the price per 100 g of each size.

(b) Which is the better buy for Jasmine? Finish the sentence.

The better buy for Jasmine is _____ because _____

(c) Give another reason (not price) that could change which size is the best buy for someone.

5. Using a Unit Price to Find a Cost

Once you know the unit price, you can find the cost of **any amount**:

Cost = unit price × amount	(the amount must be in the same unit as the unit price)

This is the same as solving a proportion. For \$2.36 per 100 g, the cost of 350 g is $\dfrac{\$2.36}{100\text{ g}} = \dfrac{x}{350\text{ g}}$, so $x = 3.5 \times \$2.36 = \$8.26$.

Example 9

(a) Kai works at a deli. Black Forest ham is \$2.36 per 100 g. Complete his price chart.

Amount	Work	Cost
100 g		\$2.36
50 g		
150 g		
350 g		
1 kg		

(b) A 400 g bag of trail mix costs \$5.60. At the same unit price, how much would 1.5 kg cost?

(c) A box of laminate flooring covers 2.4 m^2 (square metres) and costs \$67.20. Malik's bedroom floor is a rectangle 3 m by 4 m.

(i) Find the price per m^2.

(ii) Find the area of the floor.

(iii) How many boxes does Malik need, and what do they cost?

Lesson 7: Percents Refresher: Percents, Decimals and Percent of an Amount

What a percent means, switching between percents, decimals and fractions, quick percents in your head, and finding a percent of an amount or what percent one number is of another.

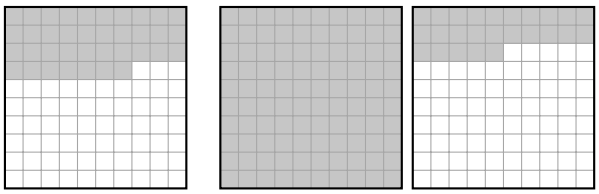
1. What Is a Percent?

Percent means "out of 100". The sign % stands for "per hundred". So 37% means 37 out of every 100.

A 10×10 grid has 100 small squares, so each square is 1%. The whole grid is **100%**. More than one whole grid is more than 100%.

Percent = per hundred.	$37\% = \dfrac{37}{100} = 0.37$	The whole = 100%.

Example 1



Grid A Grid B (two grids)

(a) What percent of Grid A is shaded? Write it three ways.

Percent _____ Fraction _____ Decimal _____

(b) What percent of Grid A is *not* shaded? _____

(c) Each grid in B is one whole. What percent is shaded? _____

2. Percents, Decimals and Fractions

A percent, a decimal and a fraction can all name the same amount. To switch:

Percent → decimal: divide by 100 (move the decimal point **2 places left**). $48\% \to 0.48$
Decimal → percent: multiply by 100 (move the decimal point **2 places right**). $0.3 \to 30\%$
Fraction → percent: divide top by bottom, then multiply by 100. $\frac{3}{4} = 3 \div 4 = 0.75 = 75\%$
Percent → fraction: write it over 100, then simplify. $60\% = \frac{60}{100} = \frac{3}{5}$

If there is no decimal point, it is at the end of the number: $7\% = 7.\% \to 0.07$. Fill empty places with zeros.

Watch out. 7% is 0.07, not 0.7. A one-digit percent needs a zero after the decimal point. Check: 7% is less than 10%, and $10\% = 0.10$, so the decimal must be less than 0.10.

Example 2

Write each percent as a decimal, and each decimal as a percent.

(a) $48\% =$ _____ (b) $7\% =$ _____ (c) $7.5\% =$ _____

(d) $125\% =$ _____ (e) $0.5\% =$ _____ (f) $300\% =$ _____

(g) $0.82 =$ _____ (h) $0.06 =$ _____ (i) $1.4 =$ _____

(j) $0.025 =$ _____ (k) $0.004 =$ _____ (l) $2 =$ _____

Calculator. For a fraction, type the top, [÷], the bottom, then [ENTER]: $7 \div 8 = 0.875$. If you type the fraction with [A b/c] instead, the screen shows a fraction ($7 \lrcorner 8$). Press [2nd] [F<>D] to change it to a decimal.

Example 3

(a) Write each fraction as a percent.
$\frac{2}{5}$ $\frac{7}{8}$ $\frac{1}{3}$ (to the nearest tenth of a percent)

(b) Write each percent as a fraction in simplest form.
15% 240%

3. Quick Percents in Your Head

Some percents are easy to find without a calculator. These are called **benchmarks**. Use them to estimate, and to check that a calculator answer makes sense.

100%	the whole amount	**10%**	divide by 10 (move the decimal **1** place left)
50%	half: divide by 2	**5%**	half of 10%
25%	a quarter: divide by 4 (or half of 50%)	**1%**	divide by 100 (move the decimal **2** places left)

Build others: $15\% = 10\% + 5\%$ $20\% = 2 \times 10\%$ $30\% = 3 \times 10\%$ $75\% = 50\% + 25\%$ $2\% = 2 \times 1\%$

Example 4

Complete the table without a calculator.

Amount	10%	5%	15%	20%	50%	25%	1%
\$60.00							
\$84.20							

Tipping. In Canada, people usually leave a **tip** of 15% to 20% of the bill at a sit-down restaurant, and often for haircuts, taxis and delivery. A tip is a percent of an amount.

Example 5

Noor and her brother have supper at a restaurant. The bill is \$47.60.

(a) Estimate a 15% tip in your head.

(b) Find a 20% tip exactly, then the total they pay.

4. Finding a Percent of an Amount

In math, **"of" means multiply**. To find a percent of an amount, change the percent to a decimal and multiply.

Percent of an amount:	part = (percent as a decimal) × amount	35% of 260 = 0.35 × 260 = 91

You can also use a proportion: $\dfrac{35}{100} = \dfrac{x}{260}$, then cross-multiply. Both ways give the same answer.

Watch out. More than 100% of an amount is *more* than the amount (150% of 40 is 60). Less than 1% of an amount is very small (0.5% of 40 is 0.2). Estimate first so you can catch a misplaced decimal point.

Calculator. Type 0.35 [×] 260 [ENTER]. On the TI-30XIIS you can also type 35 [2nd] [%], which divides by 100 for you, then [×] 260 [ENTER]. Money: round to the cent only at the end.

Example 6

Calculate. Estimate first.

(a) 35% of 260 (b) 8% of \$54.75 (c) 140% of 85

(d) 0.4% of 3200 (e) 12.5% of 96 (f) 250% of \$18

5. What Percent Is One Number of Another?

To find what percent a part is of a whole, divide the part by the whole, then multiply by 100. The **whole** is the number that comes after "of".

Percent $= \dfrac{\text{part}}{\text{whole}} \times 100$	18 is what percent of 72? $18 \div 72 = 0.25 = 25\%$

If the part is bigger than the whole, the answer is more than 100%. If the division does not come out evenly, round the percent as asked (often to the nearest tenth of a percent).

Example 7

Find each percent. Round to the nearest tenth of a percent if needed.

(a) 18 of 72 (b) 9 of 40 (c) 7 of 9 (d) 60 of 48

Example 8

(a) Priya got 38 out of 45 on her first test and 22 out of 25 on her second test. On which test did she do better?

(b) Marco left a \$12.00 tip on a \$64.00 bill. What percent tip is that, to the nearest tenth of a percent? Is it more or less than 15%?

Lesson 8: Setting a Price: Markup and Sales Tax

How a store turns what it paid into a price tag, and what you really pay at the till once tax is added.

1. Cost, Markup and Retail Price

A store does not make the things it sells. It buys them from a supplier, called a **wholesaler**. The price the store pays is the **wholesale price** (or **cost**).

The store then sells the item for more, so it can pay rent and wages and make a profit. The extra amount it adds is the **markup**. The price on the tag is the **retail price** (or **selling price**).

The markup is usually a percent of the cost, called the **markup rate**. A markup of 40% means the store adds 40% of what it paid.

Markup and retail price

markup = cost × markup rate retail price = cost + markup

Shortcut: retail price = cost × (1 + markup rate) e.g. a 40% markup: cost × 1.40

The shortcut works because the store charges 100% of the cost *plus* 40% more, which is 140% of the cost, and 140% = 1.40.

Watch out. Change the percent to a decimal first: 40% = 0.40, 5% = 0.05 (not 0.5), 180% = 1.80. A markup can be more than 100%. Then the retail price is more than double the cost.

Example 1

A bike shop pays $48.00 for a helmet. It uses a markup of 40%.

(a) Find the markup in dollars.

(b) Find the retail price. Check with the shortcut.

Example 2

(a) A phone accessory kiosk pays $6.50 for a phone case. Its markup is 180%. Find the markup and the retail price.

(b) A school café buys a box of 24 granola bars for $21.60. It marks them up 150%. What does it charge for one bar?

2. Finding the Markup Percent

Sometimes you know what the store paid and what it charges, and you want the markup rate. The markup is an **increase** from the cost, so this is a **percent increase**: compare the increase with the *starting* amount.

Percent increase $\text{percent increase} = \dfrac{\text{new} - \text{original}}{\text{original}} \times 100$

Markup percent $\text{markup percent} = \dfrac{\text{retail} - \text{cost}}{\text{cost}} \times 100$

Watch out. Always divide by the *original* amount (for markup, the cost), not by the new price.

Calculator. For a cost of \$32 and a retail price of \$56: [(] 56 [−] 32 [)] [÷] 32 [×] 100 [ENTER] gives 75. Use the brackets so the subtraction is done first.

Example 3

A skate shop buys a skateboard deck for \$32.00 and sells it for \$56.00. What is the markup percent?

Example 4

Prices go up for many reasons, not only markup.

(a) A mechanic raises her hourly rate from \$96 to \$108. What is the percent increase?

(b) A pizza place raises the price of a \$17.50 pizza by 8%. What is the new price?

3. Sales Tax: GST, PST and HST

When you buy something, the government adds **sales tax** to the retail price. There are three kinds in Canada:

- **GST** (Goods and Services Tax): 5%, charged by the federal government everywhere in Canada.
- **PST** (Provincial Sales Tax): added by some provinces on top of the GST.
- **HST** (Harmonized Sales Tax): in some provinces, the GST and the provincial tax are joined into one tax.

Alberta has GST only, so in Calgary you pay 5%. Some things, like most basic groceries (milk, bread, fresh vegetables), have no GST at all.

Province	Tax	Total	Multiply by
Alberta, territories	5% GST	5%	1.05
British Columbia	5% GST + 7% PST		
Saskatchewan	5% GST + 6% PST		
Ontario	13% HST		
Nova Scotia	14% HST		

Rates as of 2026. Tax rates can change, so check before you use them in real life.

Complete the table. The "multiply by" number works just like the markup shortcut: 100% of the price plus the tax.

Sales tax

tax = retail price × tax rate total = retail price + tax

Shortcut in Alberta: total = retail price × 1.05

Round each tax amount and each total to the nearest cent.

Estimate. To estimate 5%, find 10% (move the decimal one place left), then take half. 10% of \$54.99 is about \$5.50, so the GST is about \$2.75.

Example 5

A hoodie at a Calgary store is $54.99. Estimate the GST, then find the GST and the total cost.

Example 6

Liam buys headphones for $129.99 in Kamloops, BC (5% GST and 7% PST). Find each tax and the total.

Why work out each tax? A store must keep track of how much GST and how much PST it collects, so receipts show each tax on its own line. The shortcut is quicker when you only need the total. Once in a while the two ways differ by 1 cent because of rounding. That is fine.

Example 7

A game console costs $499.99 in Calgary and $499.99 in Toronto (13% HST). How much more does it cost in Toronto after tax?

4. Markup, Then Tax

Many problems put the two ideas together. Do them in the order they happen in a store:

Step 1: Find the retail price (cost + markup). This is the price on the tag, so round it to the cent.
Step 2: Find the tax on the *retail* price.
Step 3: Add to get the total the customer pays.

Watch out. The tax is on the retail price, not on the cost. The customer never sees the cost.

Example 8

A work-wear store in Red Deer buys steel-toed boots for $85.00 a pair. It marks them up 60%. What does a customer pay, with GST?

Example 9

Ava shops at an outdoor store in Calgary. Complete her receipt. She pays with a $100 gift card. How much is left on the gift card?

Bow Valley Outfitters Calgary, AB			
Item	**Qty**	**Price each**	**Amount**
Wool hiking socks	2	$12.99	
Toque	1	$24.50	
Water bottle	1	$18.75	
		Subtotal	
		GST 5%	
		Total	
		Gift card	$100.00
		Left on card	

Lesson 9: On Sale! Discounts and Sale Prices

Finding the discount and the sale price, the percent decrease, tax on a sale price, and deciding which sale is really the best deal.

1. Discount and Sale Price

The **regular price** (or **original price**) is the price before a sale. The **discount** is the amount of money you save. The **discount rate** is the percent off. The **sale price** is what you pay. Stores also call a discount a **markdown** (the opposite of a markup).

Discount = regular price × discount rate **Sale price** = regular price − discount

Shortcut: Sale price = regular price × (1 − discount rate). If you save 30%, you pay 100% − 30% = 70%, so multiply by 0.70.

Watch out. Change the percent to a decimal first: 30% = 0.30, not 30. For the shortcut, multiply by the part you *pay*: 30% off means × 0.70, not × 0.30.

Example 1

A winter jacket at a Calgary mall is regularly \$139.99. It is on sale for 30% off.

(a) Estimate the discount and the sale price.

(b) Calculate the discount.

(c) Calculate the sale price.

Example 2

(a) Complete the table.

Percent off	10%	25%	40%	15%	5%	60%
Percent you pay						
Multiply by						

Use the shortcut to find each sale price.

(b) Running shoes, regularly \$124.50, 40% off

(c) A phone case, regularly \$18.99, 15% off

Calculator. For 40% off \$124.50, press 124.50 [×] 0.6 [ENTER]. Round to the cent only at the end.

Example 3

(a) A carpet cleaning company charges \$89.00 per room. If you book 4 or more rooms, you get 12% off the whole bill. What do 5 rooms cost (before tax)?

(b) During "Tool Days", a hardware store has hand tools 20% off and power tools 15% off. Ali buys a hammer (\$24.99) and a drill (\$149.00). What is his total before tax?

2. Percent Decrease: How Much Off Was It?

Sometimes you know the regular price and the sale price, and you want the percent off. This is a **percent decrease**. It works just like percent increase from Lesson 8, but the price goes down.

$$\textbf{Percent decrease} = \frac{\text{amount of decrease}}{\text{original price}} \times 100 \qquad \text{where amount of decrease} = \text{original price} - \text{new price}$$

Watch out. Always divide by the **original** (regular) price, not the sale price. The percent is "out of" what it used to cost.

Example 4

Find the percent decrease.

(a) A gaming headset goes from \$80.00 to \$60.00.

(b) A used truck is listed for \$18 500 but sells for \$15 900. (Round to the nearest tenth of a percent.)

3. Sale Price, Then Tax

Sales tax is charged on the **sale price**, the price you actually pay, not the regular price. So take the discount off first, then add the tax. In Alberta that is 5% GST, so multiply by 1.05. In other provinces use the rate from Lesson 8 (BC: 5% GST + 7% PST = 12%; Ontario: 13% HST; Nova Scotia: 14% HST; rates can change).

Total cost = regular price $\times (1 - \text{discount rate}) \times (1 + \text{tax rate})$

Example: 35% off, then 5% GST: regular price $\times\, 0.65 \times 1.05$

Calculator. You can do it in one line: 84 [×] 0.65 [×] 1.05 [ENTER]. Then round to the cent.

Example 5

In Calgary, a blender is regularly \$84.00. It is on sale for 35% off. GST is 5%.

(a) Find the sale price.

(b) Find the GST.

(c) Find the total cost.

Example 6

On a trip to Kelowna, BC, Priya buys a tent that is regularly \$260.00. It is 25% off. BC charges 5% GST and 7% PST.

(a) What is the sale price?

(b) What is her total cost with taxes?

(c) What would she pay with taxes at the regular price? How much does she save in total?

(d) What percent of the regular total does she save?

4. Comparing Promotions

A **promotion** is anything a store does to get you to buy: a percent off, dollars off, "buy one, get one" (BOGO), "buy 2, get 1 free", and so on. To compare promotions, work out what you would *really* pay, or the percent you would save.

$$\textbf{Percent saved} = \frac{\text{amount saved}}{\text{regular price of everything you take home}} \times 100$$

Is it a deal for you? The biggest percent is not always the best choice. Ask: Do I need that many? Will it spoil or go out of style? Do I have room to store it? Is the store close, with a good return policy?

Example 7

One store offers **30% OFF** and another offers **$25 OFF** any item. Which promotion gives the lower price?

(a) Jeans, regularly $60.00 (b) A jacket, regularly $120.00 (c) Boots, regularly $85.00

Example 8

T-shirts are regularly $22.00 each at two stores.

Store A: **Buy one, get the second half price** Store B: **Buy 2, get 1 free**

(a) At Store A, find the cost of 2 shirts and the percent saved.

(b) At Store B, find the cost of 3 shirts and the percent saved.

(c) Which is the better deal? Give a reason that is about more than the price.

5. Discount on a Discount, and "Is It Really a Deal?"

Some sales take an **extra** percent off a price that is already on sale. The second discount is taken from the *lower* sale price, so the two percents do **not** add. Multiply by each "percent you pay", one after the other.

Two discounts: final price = regular price $\times (1 - \text{first rate}) \times (1 - \text{second rate})$

20% off, then an extra 10% off: $0.80 \times 0.90 = 0.72$. You pay 72%, so you save 28%, **not** 30%.

Example 9

Hiking boots are regularly $150.00. They are 20% off, and on Saturday you get an extra 10% off at the till.

(a) Find the final price.

(b) What is the total percent off? Is it the same as 30% off?

A big "% off" sign does not always mean a good price. Some stores **raise the price** before a sale, so the sale price is close to, or even more than, the old price. Compare the sale price with what the item used to cost.

Example 10

A TV was $500.00 in October. The week before a "Black Friday" sale, the store raised its price by 20%. Then it advertised **15% OFF!**

(a) What was the raised price?

(b) What is the sale price?

(c) Is it really a deal? Find the percent change from the October price.

Lesson 10: Currency Exchange: Converting, and Buy vs Sell Rates

Changing Canadian dollars into other money and back, estimating foreign prices in your head, and why the bank's two rates cost you money.

1. What an Exchange Rate Means

Each country has its own money, called its **currency**. Banks write each currency with a three-letter **code**.

Currency	Code	Used in
Canadian dollar	CAD	Canada
US dollar	USD	United States
euro	EUR	France, Germany, Italy, Spain and many other European countries
British pound	GBP	United Kingdom (England, Scotland, Wales, Northern Ireland)
Mexican peso	MXN	Mexico
Japanese yen	JPY	Japan

An **exchange rate** tells you how much of one currency you get for **1 unit** of another currency. It can be written two ways:

1 USD = 1.38 CAD means one US dollar costs $1.38 in Canadian money.
1 CAD = 0.725 USD means one Canadian dollar is worth about 72 US cents.
If the number after the equals sign is **bigger than 1**, then 1 unit of the first currency is worth **more** than 1 unit of the second.

Note. The exchange rates in this booklet are examples. Real rates change every day; banks post them on their websites.

Example 1

Use 1 USD = 1.38 CAD.

(a) Which is worth more, one US dollar or one Canadian dollar? _____

(b) An exchange rate is a unit rate, so you can scale it up. Complete the table.

USD	$1.00	$10.00	$50.00	$200.00
CAD				

(c) In Japan, 1 CAD = 108 JPY. Is one yen worth more or less than one Canadian cent? Explain.

2. Converting Between Currencies

Converting money is a proportion problem: the rate is a unit rate (for 1 unit), and you scale it up. Look at which currency the rate starts with.

Step 1. Write the rate as 1 [one currency] = rate [other currency].
Step 2. If the rate starts with **the currency of the amount you have**, **multiply**: amount × rate.
If it starts with **the other currency**, **divide**: amount ÷ rate.
Step 3. Round money to the cent at the end (yen to the whole yen). Write the currency code.

Watch out. Before you press ENTER, ask: should my number get bigger or smaller? A US dollar is worth more than a Canadian dollar, so the same price is a **bigger number in CAD** than in USD. If your CAD answer is smaller than the US price, you multiplied when you should have divided (or the other way around).

Calculator. To change $350.00 CAD at 1 CAD = 0.725 USD: 350 × 0.725 ENTER shows 253.75.

Example 2

(a) Hamza changes $350.00 CAD into US dollars. The rate is 1 CAD = 0.725 USD. How many US dollars does he get?

(b) A hotel in Great Falls, Montana, costs $164.00 USD a night. The rate is 1 USD = 1.38 CAD. What is the cost in Canadian dollars?

Sometimes only one rate is posted, and it starts with the currency you *want*. Then divide, or set up a proportion and cross-multiply.

Example 3

A travel website shows only one rate: 1 EUR = 1.55 CAD.

(a) Sofia has $450.00 CAD. How many euros can she get?

(b) A train ticket in Italy costs 38.40 EUR. What is that in CAD?

(c) A shop in Tokyo posts only 1 CAD = 108 JPY. A bento lunch box costs 1250 JPY. What is that in CAD?

Banks and travel sites show rates for many currencies in a **rate table**. Choose the column that starts with the currency you **have**, then multiply.

Example exchange rates		
Currency	**1 unit = _____ CAD**	**1 CAD = _____ units**
US dollar (USD)	1.38	0.725
euro (EUR)	1.55	0.645
British pound (GBP)	1.82	0.549
Mexican peso (MXN)	0.075	13.3
Japanese yen (JPY)	0.0093	108

Example 4

Use the rate table above. Show your multiplication.

(a) Change $250.00 CAD into Japanese yen.

(b) Lena comes home from Mexico with 1800 MXN. What are they worth in CAD?

(c) A theatre ticket in London costs 46.50 GBP. What is that in CAD?

(d) Change $80.00 CAD into euros.

Note. Yen have no cents, so round yen to the whole yen. A peso is worth only a few Canadian cents, so a price of hundreds of pesos is not very much money.

3. Estimating Canadian Prices in Your Head

When you shop in the US, the price tags are in US dollars. You usually have no calculator out, but you still need to know what things really cost. With 1 USD $\approx$ 1.38 CAD, each US dollar costs a bit more than $1\frac{1}{3}$ Canadian dollars.

Estimating a US price in CAD (when 1 USD is about 1.35 to 1.40 CAD)
Step 1. Round the US price to a friendly number. **Step 2.** Find about a third of it ($\div 3$). **Step 3.** Add it on.
Or multiply by 1.4 in your head ($\times 1$ plus $\times 0.4$). This gives a slightly high estimate, which is safe for a budget.

Why it matters. Estimating lets you keep to your budget, tell whether a "sale" is really cheaper than at home, and avoid a surprise when your credit card bill arrives in Canadian dollars. It also helps you catch a calculator mistake. The same idea works for other currencies: with 1 EUR $\approx$ 1.55 CAD, add about half.

Example 5

Use 1 USD = 1.38 CAD. Estimate first, then find the exact cost.

(a) A ball cap costs \$24.00 USD.

(b) Running shoes cost \$119.99 USD.

(c) Kayden's calculator says a \$45.00 USD hoodie costs \$32.61 CAD. Is that reasonable? What went wrong?

4. Buying and Selling Rates

A bank or currency exchange does not use one rate. It posts **two rates**, written from the **bank's point of view**, in CAD for 1 unit of foreign money:

- The **selling rate** ("we sell") is what the bank charges when it **sells you** foreign cash, for example before a trip. It is the **higher** rate.
- The **buying rate** ("we buy") is what the bank pays when it **buys** foreign cash back from you, for example after a trip. It is the **lower** rate.

The gap between the two rates is how the bank makes money. So if you change money and then change it back, you get back **less** than you started with.

Getting foreign cash (you buy, the bank sells): use the **selling rate** (higher).
Changing it back to CAD (you sell, the bank buys): use the **buying rate** (lower).
Either way the bank's rate is the worse one for you: you pay more, and you get less.

Example Bank: foreign cash (CAD for 1 unit)		
Currency	**We buy**	**We sell**
US dollar (USD)	1.34	1.42
euro (EUR)	1.50	1.60
Mexican peso (MXN)	0.070	0.080

Example 6

Owen is going to Phoenix. Use the Example Bank board.

(a) Owen buys \$600.00 USD in cash. Which rate is used? How much does it cost him in CAD?

(b) His sister has \$500.00 CAD to change. How many US dollars does she get?

(c) Owen comes home with \$140.00 USD left and sells it back to the bank. How much CAD does he get?

(d) How much did those \$140.00 USD cost him before the trip? How much did he lose by changing them back?

5. Fees, and Shopping Across the Border

Some exchange counters also charge a **fee** (a flat amount, such as \$9.95) or a **commission** (a percent of the amount). A good-looking rate with a fee can cost more than a worse rate with no fee, so always compare the **total** cost.

When you buy from a US website with a credit card, the card company converts the price and usually adds a **foreign transaction fee**, often about 2.5%. To compare with a Canadian store, compare the total in CAD with the store's price *plus GST*.

Total cost in CAD = (price in foreign money × rate) + flat fee or (price × rate) × (1 + commission rate)

Example 7

Lucia needs 250 EUR in cash for a trip to Spain. Complete the table to find the cheapest place to buy them.

Place	**Rate and fee**	**Working**	**Total (CAD)**
Bank	1 EUR = 1.60 CAD, no fee		
Airport kiosk	1 EUR = 1.57 CAD, plus \$9.95 fee		
Mall exchange	1 EUR = 1.55 CAD, plus 4% commission		

Example 8

Tyler wants a hoodie. A US website sells it for \$52.00 USD plus \$11.50 USD shipping. His credit card uses 1 USD = 1.38 CAD and adds a 2.5% foreign transaction fee. A store in Calgary sells the same hoodie for \$79.99 plus 5% GST. (Ignore any charges at the border.)

(a) What is the total cost of the online hoodie in CAD?

(b) What is the total cost at the Calgary store?

(c) Which is the better deal? Give one reason that is not about price.

Lesson 11: Unit 1 Review

Every rule from the unit on one page, worked examples that mix the topics, and a practice unit test.

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1. The Unit on One Page

Use this page to study. Each row is one skill: what to do, and a short example.

Skill	What to do	Example
Ratio	Compares two amounts in the **same units**. Write 3 : 5, 3 to 5 or $\frac{3}{5}$. Change to the same units first, then divide both terms by the same number.	45 min : 2 h = 45 : 120 = 3 : 8
Sharing in a ratio	Add the parts. Divide the total by the number of parts to get one part. Multiply.	\$120 in 3 : 5: 8 parts, \$15 a part, so \$45 and \$75
Proportion	Two equal ratios. Put matching units in matching places. Scale up or down, or cross-multiply.	$\frac{x}{6} = \frac{10}{15}$: $15x = 60$, so $x = 4$
Rate and unit rate	A rate compares **different units**. A unit rate is "per 1": divide by the second amount.	240 km in 3 h = 80 km/h
Fuel economy	L/100 km = litres ÷ km × 100. A smaller number uses less fuel.	36 L for 450 km: 36 ÷ 450 × 100 = 8 L/100 km
Unit price	unit price = price ÷ quantity. Per 100 g: price ÷ grams × 100. Use the same unit for every size. Keep an extra decimal when comparing.	\$3.49 for 750 g: 3.49 ÷ 750 × 100 = \$0.465 per 100 g
Percent of an amount	Change the percent to a decimal (7.5% = 0.075), then multiply.	15% of \$42 = 0.15 × 42 = \$6.30
Percent increase or decrease	change ÷ original × 100. Always divide by the **original** (old) price.	\$40 to \$50: 10 ÷ 40 × 100 = 25% increase
Markup	markup = cost × rate; retail = cost + markup = cost × (1 + rate). Markup % = markup ÷ cost × 100.	\$20 cost, 40% markup: 20 × 1.40 = \$28.00
Sales tax	tax = price × rate; total = price × (1 + rate). Alberta: GST 5%, so ×1.05. Other provinces add PST or use HST.	\$28.00 in Alberta: 28 × 1.05 = \$29.40
Discount	discount = original × rate; sale = original × (1 − rate). Tax goes on the **sale** price.	\$60, 25% off: 60 × 0.75 = \$45.00
Currency exchange	Write the rate as 1 [currency] = rate. If it starts with the currency you **have**, multiply; if it starts with the currency you **want**, divide. The bank **sells** you foreign cash at its higher rate and **buys** it back at its lower rate.	\$50.00 USD at 1 USD = 1.38 CAD: 50 × 1.38 = \$69.00 CAD

Remember. Estimate first, so you can spot a wrong calculator answer. Round money to the cent only at the **end**. A best buy is about price **and** other things: spoiling, storage, quality, how much you need. Two discounts in a row are **not** added: take the second one off the new, lower price.

2. Ratios, Proportions and Rates

Example 1

(a) Write each ratio in simplest form.

(i) 36 : 48 _____

(ii) 90 cm : 1.5 m _____

(b) Priya and Jordan shovelled snow for $250. Priya worked 6 h and Jordan worked 4 h, so they split the money in the ratio 6 : 4. How much does each person get?

Example 2

(a) A café's iced tea uses 3 cups of tea concentrate for every 8 cups of water. How much concentrate goes with 20 cups of water?

(b) A road map of southern Alberta has a scale of 1 cm : 12 km. Calgary and Drumheller are 11.5 cm apart on the map. About how far apart are they really?

Example 3

Mateo's truck used 54 L of gas to drive 600 km.

(a) What is the truck's fuel economy in L/100 km?

(b) Mateo drives 970 km from Calgary to Vancouver. Gas costs $1.42/L. Estimate the cost of gas, then calculate it.

3. Finding the Best Buy

Example 4

A Calgary grocery store sells ground coffee in three sizes.

Size	340 g	925 g	1.36 kg
Price	$8.99	$19.49	$27.99
Unit price (per 100 g)			

(a) Complete the table. Which size is the best buy?

(b) Give one reason that is **not** about price for buying the 340 g bag.

(c) Four 340 g bags hold 1.36 kg. How much do you save by buying one 1.36 kg bag instead of four 340 g bags?

4. Markup, Tax and Sales

Example 5

A bike shop pays a wholesale price of $420.00 for a bike and marks it up 45%.

(a) What is the retail price?

(b) What does the bike cost with tax in Alberta (5% GST)?

(c) What would it cost in Ontario (13% HST)?

(d) The shop buys helmets for $32.00 and sells them for $56.00. What is the markup percent?

Example 6

A winter coat that regularly sells for $149.00 is on sale for 35% off.

(a) Find the discount and the sale price.

(b) Find the GST and the total you pay in Calgary.

(c) Another store sells the same coat for $104.30 (before tax). What percent decrease is that from $149.00?

Example 7

Runners cost $90.00 at two stores. **Store A**: 25% off, then an extra 10% off at the till. **Store B**: $30 off.

(a) Find the price at each store before tax. Which is cheaper?

(b) Is "25% off, then 10% off" the same as 35% off? Explain.

5. Currency Exchange

Watch out. On a bank's board, "We sell" is the rate when *you buy* foreign cash (higher), and "We buy" is the rate when *you sell* it back (lower). Rates in this booklet are examples.

Example 8

A bank in Calgary shows these rates, in Canadian dollars for 1 unit of each currency.

Currency	We buy	We sell
US dollar (USD)	1.34	1.42
Euro (EUR)	1.46	1.56
Mexican peso (MXN)	0.072	0.081

(a) Kayla buys $300 USD in cash for a trip to Montana. How much does it cost her in Canadian dollars?

(b) She comes home with $40 USD and sells it back to the bank. How much does she get? How much did she lose on that $40 USD?

(c) In Montana a hoodie costs $45.00 USD. Estimate the cost in Canadian dollars in your head, then find it exactly using 1 USD = 1.38 CAD. Why does estimating help when you shop?

(d) Her friend buys 3000 Mexican pesos at this bank for a trip to Cancun. How much does that cost in Canadian dollars?